HKUST – Department of Computer Science and Engineering
COMP 2711: Discrete Math Tools for Computer Science

Fall 2025 Final Examination

Date: 15 Dec 2025 Time: 12:30-15:00

Name: _____	Student ID: _____
Email: _____	Lecture: _____

Instructions

- This is a closed book exam. It consists of **11** questions. You are allowed to bring an A4 double-sided, handwritten, cheat sheet.
- Please write your name, student ID, email, and lecture section in the space provided at the top of this page.
- Please sign the honor code statement on page 2.
- Answer all questions within the space provided on the examination paper. You may also use the back of the pages. Each question is on a separate page. This is for clarity and is not meant to imply that each question requires a full-page answer. Some can be answered using only a few lines.
- Please use **pen** to answer the questions. Answers written in pencil will not be accepted for paper-checking.
- Solutions can be written in terms of binomial coefficients, factorials, and the $C(n,k)$, $P(n,k)$ notation. For example, you can write $\binom{5}{3}+\binom{4}{2}$ instead of 16. Avoid using nonstandard notations such as ${}_nP_k$ or ${}_nC_k$.

Problems	1	2	3	4	5	6	7	8	9	10	11	Total
Points	10	12	9	9	6	8	9	5	10	10	12	100
Score												

As part of HKUST's introduction of an honor code, the HKUST Senate has recommended that all students be asked to sign a brief declaration printed on examination answer books that their answers are their own work, and that they are aware of the regulations relating to academic integrity. Following this, please read and sign the declaration below.

I declare that the answers submitted for this examination are my own work.

I understand that sanctions will be imposed, if I am found to have violated the University regulations governing academic integrity.

Student's Name: _____

Student's Signature: _____

Problem 1: Logic [10 points]

(a) **[3 points]** Without using truth table prove that:

$$[(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r)] \equiv p \wedge q$$

(b) **[3 points]** Let p, q, r, s, t denote five propositions. You are given the following premises:

1.$p \wedge q$
2. $p \rightarrow (r \wedge q)$
3. $r \rightarrow (s \vee t)$
4. $\neg s$

Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion "t" is true.

(c) **[4 points, 2 pts each]** Lets the domain, be the students of COMP2711. Consider the following predicates:

$s(x)$: Student x is a senior.
$m(x)$: Student x is a mathematics major.
$c(x)$: Student x is a comp major
$Compare(x, y)$: Student x gets a higher grade than student y
Write each of the following statement in terms of quantifiers, logical operators and the predicates above.

(i) Every senior in the class is majoring either in comp or mathematics.
(ii) The highest grade belongs to a comp major student.

Answer: (a)

$(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r)$

$\equiv \neg(\neg p \vee \neg q) \vee (p \wedge q \wedge r)$ (implication equivalence)

$\equiv (p \wedge q) \vee (p \wedge q \wedge r)$ (De Morgan's law)

$\equiv (p \wedge q) \vee \big((p \wedge q) \wedge r\big)$ (associativity)

$\equiv (p \wedge q) \wedge \big(1 \vee r\big)$ (distributivity: $A \vee (A \wedge B) \equiv A \wedge (1 \vee B)$)

$\equiv (p \wedge q) \wedge 1$ (since $1 \vee r \equiv 1$)

$\equiv p \wedge q$ (identity law for $\wedge$)

Thus,

$$\boxed{[(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r)] \equiv p \wedge q}$$

(Note: the proposition r cancels out completely and does not affect the equivalence.)

(b)

1. $p \wedge q$	(given)
2. p	(1, simplification)
3. $p \rightarrow (r \wedge q)$	(given)
4. $r \wedge q$	(2, 3, modus ponens)
5. r	(4, simplification)
6. $r \rightarrow (s \vee t)$	(given)
7. $s \vee t$	(5, 6, modus ponens)
8. $\neg s$	(given)
9. t	(7, 8, disjunctive syllogism)

Thus, the premises logically imply t.

(c) (i) $\boxed{\forall x\big(s(x) \rightarrow \big(c(x) \vee m(x)\big)\big)}$

(ii) $\boxed{\exists y\left(c(y) \wedge \forall x\left(x \neq y \rightarrow \neg\,\mathrm{Compare}(x, y)\right)\right)}$

Problem 2: Set and Combinatorics [12 points]

(a) **[3 points , 1 pt each]** Determine whether each of these sets is **finite**, **countably infinite**, or **uncountable**. **No argument is needed**.

(i) The set of binary strings of infinite length with no two identical consecutive bits (i.e., no "00" and no "11" anywhere in the sequence).

(ii) The set of ternary strings (i.e., they are made of $\{0, 1, 2\}$) of infinite length.

(iii) The set of ternary strings of infinite length such that they contain "012".

(b) **[5 points]** Consider $A = \{1, 2, 3, 4, 5\}$. Imagine picking 17 different subsets from A at random. Prove that at least two of them are disjoint.

(c) **[4 points]** Use combinatorial argument to prove:

$$\binom{n}{k} P(k, r) = \binom{n-r}{k-r} P(n, r)$$

Answer: (a) (i) Countable infinite and Finite are both accepted here. based on the definition, the correct answer is finite, because there are only 2 such strings. However if the phrasing of the question was set of arbitrary length strings ... then the answer was countable infinite. As it was not discussed clearly, we accept both Countable Infinite and Finite answers.

(ii) Uncountable

(iii) Uncountable

(b) The power set of A has $2^5 = 32$ subsets. For each subset $S \subseteq A$, its complement S^c is disjoint with S (i.e., $S \cap S^c = \emptyset$).

Also the complement of any subset is unique (i.e., there is a bijection between any subset and its complement).

Thus, these 32 subsets can be paired into 16 groups of (subsets and their complement). Based on pigeonhole principle, if we choose 17 subsets, two of them will be from the same pair which means they are disjoint.

(c) From a group of n distinct people, choose an ordered committee of r people (positions matter) and also choose an unordered group of k people who will be the full selection committee (the r ordered members are part of these k).

Left-hand side: - First choose the k members of the full selection committee: $\binom{n}{k}$ ways. - Then choose r ordered positions (e.g., chairperson, vice-chair, etc.) from these k people: $P(k, r)$ ways. Total: $\binom{n}{k} \cdot P(k, r)$.

Right-hand side: - First choose the r people who will hold the r ordered positions, in order: $P(n, r)$ ways. - Now we still need to choose the remaining $k-r$ members of the full (unordered) committee. These must come from the remaining $n-r$ people (excluding the r already chosen for the ordered roles): $\binom{n-r}{k-r}$ ways. Total: $\binom{n-r}{k-r} \cdot P(n, r)$.
Both sides count the same thing, so

$$\binom{n}{k} \cdot P(k, r) \;=\; \binom{n-r}{k-r} \cdot P(n, r).$$

Problem 3: Probability [9 points] We have 14 volleyball players, all of different heights. All 14! permutations of rankings (tallest = rank 1 to shortest = rank 14) are equally likely. Answer the following questions and briefly explain. You do not need to calculate the final numerical answer.

(a) **[3 points]** Given A is taller than B, what is the probability that A is tallest?

(b) **[3 points]** Given A is taller than B, what is the probability that A is the third tallest?

(c) **[3 points]** Given A is taller than B and shorter than C, what is the probability that A is seventh tallest?

Answer: (a) Given $A > B$, permutations satisfying this: $14!/2$. If A is tallest, arrange others: 13! permutations.

$$\Pr(A \text{ tallest} \mid A > B) = \frac{13!}{14!/2} = \frac{2}{14} = \frac{1}{7}.$$

(b) Given $A > B$. If A is rank 3, B must be in ranks $4, \ldots, 14$: 11 choices. Then arrange remaining 12 players: 12!.

$$\Pr(A \text{ rank } 3 \mid A > B) = \frac{11 \cdot 12!}{14!/2} = \frac{11 \cdot 2}{14 \cdot 13} = \frac{11}{91}.$$

(c) Given $C > A > B$. Permutations satisfying this: $14!/6$. If A is rank 7, C in ranks $1, \ldots, 6$ (6 choices), B in ranks $8, \ldots, 14$ (7 choices), arrange remaining 11: 11!.

$$\Pr(A \text{ rank } 7 \mid C > A > B) = \frac{6 \cdot 7 \cdot 11!}{14!/6} = \frac{252}{14 \cdot 13 \cdot 12} = \frac{3}{26}.$$

Problem 4: Combinatorics [9 points]

Assume there are 4 teachers (T), 5 engineers (E), 6 doctors (D) and 7 students (S). These $4+5+6+7 = 22$ people want to stand in a single queue (a line with a well-defined front/head and back/end). Every individual is a distinct person (i.e., they are all different people)

Determine the number of ways to arrange them in the queue for each of the following conditions. **No argument is needed.** There is no need to calculate the exact numerical answer.

You are not allowed write the answer in any form of summation.

(a) **[3 points]** Every teacher stands **in front of** all doctors (i.e., the last teacher is before the first doctor in the queue).

(b) **[3 points]** Every teacher has **at least one doctor** behind (i.e., no teacher is after the last doctor). Note that *behind* does not necessarily mean the previous person.

(c) **[3 points]** The first teacher and first engineer are standing next to each other (By first teacher and first engineer we mean the earliest in the queue toward its front).

Answer: (a) $\binom{22}{12} \times 12! \times 6! \times 4!$

(b) $\binom{22}{12} \times 12! \times 6 \times 9!$

(c) $\binom{21}{13} \times 13! \times (4 \times 5 \times 2) \times 7!$

Problem 5: Conditional probability [6 points]

Suppose that there are two slot machines, one of which pays out 10% of the time and the other pays out 20% of the time. Unfortunately, you have no idea which is which. Suppose you randomly choose a machine and put in a coin. Answer the following questions: You do not need to calculate the final numerical value.

(a) **[3 points]** If you don't get paid, what is the chance that you chose the machine that pays out 20% of the time?

(b) **[3 points]** If you get paid, what is the chance that you chose the machine that pays out 20 % of the time?

Answer: Let S be the event that you choose the first slot machine, i.e. the one that pays out 10% of the time (so S^c is the event that you choose the second slot machine). Let J be the event that you get a jackpot. The information given in the problem is

- $P(S) = 0.5$
- $P(J \mid S) = 0.1$
- $P(J \mid S^c) = 0.2$

We want to find $P(S^c \mid J^c)$ and $P(S^c \mid J)$. We can find both by applying Bayes' theorem, just as in the previous problem (though this time we won't explain every detail of the calculation):

$$P(S^c \mid J^c) = \frac{P(J^c \mid S^c)P(S^c)}{P(J^c)} = \frac{P(J^c \mid S^c)P(S^c)}{P(J^c \mid S^c)P(S^c) + P(J^c \mid S)P(S)}$$

$$= \frac{0.8 \cdot 0.5}{0.8 \cdot 0.5 + 0.9 \cdot 0.5} = \boxed{0.471}$$

Similarly,

$$P(S^c \mid J) = \frac{P(J \mid S^c)P(S^c)}{P(J)} = \frac{P(J \mid S^c)P(S^c)}{P(J \mid S^c)P(S^c) + P(J \mid S)P(S)}$$

$$= \frac{0.2 \cdot 0.5}{0.2 \cdot 0.5 + 0.1 \cdot 0.5} = \boxed{0.667}$$

Question 6: Chinese Remainder Theorem [8 pts]

Use Chinese Remainder Theorem to solve the following system of congruences:

$$\begin{aligned} x &\equiv 1 \pmod 2 \\ x &\equiv 1 \pmod 3 \\ x &\equiv 3 \pmod 4 \\ x &\equiv 1 \pmod 5 \end{aligned}$$

You must show all the steps. If you just write down your answer, you will get 0 points even if the answer is correct.

Hint: Note that one of the above congruences is redundant and can be omitted.

Answer: Note that the first equation is already included in the third one. So we can simply ignore the first equation. And just consider the following congruence system :

$$\begin{aligned} x &\equiv 1 \pmod 3 \\ x &\equiv 3 \pmod 4 \\ x &\equiv 1 \pmod 5 \end{aligned}$$

We solve the system

$$\begin{aligned} x &\equiv 1 \pmod 3 \\ x &\equiv 3 \pmod 4 \\ x &\equiv 1 \pmod 5 \end{aligned}$$

using the explicit CRT formula. The moduli are pairwise coprime and $M = 3 \cdot 4 \cdot 5 = 60$.

Define $M_1 = M/3 = 20, \quad M_2 = M/4 = 15, \quad M_3 = M/5 = 12.$

Find the inverses:

- y_1 such that $20y_1 \equiv 1 \pmod 3$: $20 \equiv 2 \pmod 3$, inverse of 2 modulo 3 is 2 (because $2 \cdot 2 = 4 \equiv 1$). $\Rightarrow y_1 = 2$.
- y_2 such that $15y_2 \equiv 1 \pmod 4$: $15 \equiv 3 \pmod 4$, inverse of 3 modulo 4 is 3 (because $3 \cdot 3 = 9 \equiv 1$). $\Rightarrow y_2 = 3$.
- y_3 such that $12y_3 \equiv 1 \pmod 5$: $12 \equiv 2 \pmod 5$, inverse of 2 modulo 5 is 3 (because $2 \cdot 3 = 6 \equiv 1$). $\Rightarrow y_3 = 3$.

Now apply the formula:

$$\begin{aligned} x &= a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 \\ &= 1 \cdot 20 \cdot 2 + 3 \cdot 15 \cdot 3 + 1 \cdot 12 \cdot 3 \\ &= 40 + 135 + 36 \\ &= 211 \end{aligned}$$

Reduce modulo $M = 60$:

$$211 \equiv 31 \pmod{60} \qquad (211 - 3 \cdot 60 = 211 - 180 = 31)$$

Thus the unique solution is

$$\boxed{x \equiv 31 \pmod{60}}$$

Question 7: RSA [9 pts]

This problem is on the RSA algorithm for public key cryptogtaphy. To generate his keys, Bob uses two prime numbers $p = 7, q = 11$. So, $n = pq = 77$ and $T = (p-1)(q-1) = 60$.

(i) **[3 pts]** Write down the possible values for e if it has to be between 10 and 20. No argument is needed.

(ii) **[3 pts]** Assume that Bob chooses $e = 7$. Compute his private key d by running the extended Euclidean algorithm.

(iii) **[3 pts]** Suppose Alice wants to encrypt message $m = 3$ using Bob's public key (n, e). Note that $n = 77$, $e = 7$. Use the repeated squaring technique to calculate the ciphertext $c = m^e$.

Answer: (i) Possible e with $10 \leq e \leq 20$ and $\gcd(e, 60) = 1$:
$e = 11, 13, 17, 19$

(ii) Bob chooses $e = 7$. Compute d such that $7d \equiv 1 \pmod{60}$ using Extended Euclidean Algorithm:

$60 = 8 \cdot 7 + 4$
$7 = 1 \cdot 4 + 3$
$4 = 1 \cdot 3 + 1$
$3 = 3 \cdot 1 + 0$

Back-substitute:

$1 = 4 - 1 \cdot 3$ $3 = 7 - 1 \cdot 4 \implies 1 = 4 - 1 \cdot (7 - 1 \cdot 4) = 2 \cdot 4 - 1 \cdot 7$
$4 = 60 - 8 \cdot 7 \implies 1 = 2 \cdot (60 - 8 \cdot 7) - 1 \cdot 7 = 2 \cdot 60 - 17 \cdot 7 \implies$
$-17 \cdot 7 \equiv 1 \pmod{60} \implies d \equiv 43 \pmod{60}$ $\boxed{d = 43}$

(iii) Encrypt message $m = 3$ using Bob's public key $(n = 77, e = 7)$ by repeated squaring to compute $c = 3^7 \bmod 77$.

$$3^1 \equiv 3 \pmod{77}$$
$$3^2 \equiv 9 \pmod{77}$$
$$3^4 = (3^2)^2 \equiv 9^2 = 81 \equiv 4 \pmod{77}$$

Thus, the ciphertext is $3^7 = 3 \times 3^2 \times 3^4 \pmod{77} = 3 \times 9 \times 4 \pmod{77} = 31$

$$\boxed{c = 31}$$

Problem 8: Time Complexity [5 points]

We have two algorithms, A and B. Let $T_A(n)$ and $T_B(n)$ denote the time complexities of algorithms A and B respectively. For each case, decide whether A is faster, B is faster, or undetermined (i.e., the information provided is not enough to conclude which algorithm is faster). No explanation is needed.

(i) **[1 point]** $T_A(n) = \Theta(\sum_{i=1}^{n} n^2)$ $\quad$ $T_B(n) = \Theta(n^4)$

(ii) **[1 point]** $T_A(n) = O(\sum_{i=1}^{n} n^2)$ $\quad$ $T_B(n) = \Omega(n^{3+\epsilon})$ $(\epsilon > 0)$

(iii) **[1 point]** $T_A(n) = \Omega(\sqrt{n})$ $\quad$ $T_B(n) = O(\log n)$

(iv) **[1 point]** $T_A(n) = \Omega(n^2)$ $\quad$ $T_B(n) = \Omega(n \log^2 n)$

(v) **[1 point]** $T_A(n) = \Theta(n^3)$ $\quad$ $T_B(n) = \Theta(n^2 \log(n!))$

Answer: Using expressions like <> is not acceptable.The question is specifically asking which algorithm is faster. There can be several confusions here and the goal is to see if you understand the differences. When the growth function is larger, the algorithm is slower, when the grows function is smaller the algorithm is faster.

No arguments can be made here unless there are some mistakes.

(i) A is faster

(ii) A is faster

(iii) B is faster

(iv) undetermind.

(v) A is faster

Problem 9: Recurrence relation and Induction [10 points]

(a) **[5 points]** Solve the following recurrence relation using the expansion method:

$$T(n) = 3T(\frac{n}{4}) + n^2 \qquad T(1) = 1$$

You may assume n is of form 4^k for some positive integer k.

(b) **[5 points]** Prove your result in part (a) using induction.

Answer: (a) We solve the recurrence

$$T(n) = 3T\left(\frac{n}{4}\right) + n^2, \qquad T(1) = 1,$$

where $n = 4^k$ for some positive integer k.
Using the iteration (unrolling) method:

$$\begin{aligned} T(n) &= 3T\left(\frac{n}{4}\right) + n^2 \\ &= 3\Big[3T\left(\frac{n}{16}\right) + \left(\frac{n}{4}\right)^2\Big] + n^2 = 3^2T\left(\frac{n}{16}\right) + 3\cdot\frac{n^2}{16} + n^2 \\ &= 3^3T\left(\frac{n}{64}\right) + 3^2\cdot\frac{n^2}{16^2} + 3\cdot\frac{n^2}{16} + n^2. \end{aligned}$$

After k iterations (until the argument becomes 1, because $n/4^k = 4^k/4^k = 1$),

$$T(n) = 3^kT(1) + n^2\sum_{i=0}^{k-1} 3^i\cdot\left(\frac{1}{16}\right)^i.$$

Since $T(1) = 1$ and $n = 4^k$, we have $3^k = (4^{\log_4 3})^k = n^{\log_4 3}$. The sum is a geometric series with ratio $r = 3/16$:

$$\sum_{i=0}^{k-1} 3^i\frac{1}{16^i} = \sum_{i=0}^{k-1}\left(\frac{3}{16}\right)^i = \frac{1-(3/16)^k}{1-3/16} = \frac{1-(3/16)^k}{13/16} = \frac{16}{13}\Big(1-\left(\frac{3}{16}\right)^k\Big).$$

Thus

$$T(n) = 3^k + n^2\cdot\frac{16}{13}\Big(1-\left(\frac{3}{16}\right)^k\Big).$$

Substituting $3^k = n^{\log_4 3}$ and $(3/16)^k = 3^k/16^k = 3^k/n^2 = n^{\log_4 3 - 2}$ (or directly $(3/16)^k = 3^k/n^2$),

$$T(n) = n^{\log_4 3} + \frac{16}{13}n^2\left(1 - \frac{n^{\log_4 3}}{n^2}\right) = n^{\log_4 3} + \frac{16}{13}n^2 - \frac{16}{13}n^{\log_4 3}.$$

Combining the terms with the same exponent,

$$T(n) = \left(1 - \frac{16}{13}\right) n^{\log_4 3} + \frac{16}{13}n^2 = -\frac{3}{13}n^{\log_4 3} + \frac{16}{13}n^2 = \frac{16n^2 - 3n^{\log_4 3}}{13}.$$

Because $\log_4 3 \approx 1.584 < 2$, the dominant term is $\frac{16}{13}n^2$, so $T(n) = \Theta(n^2)$. The exact closed form is

$$\boxed{T(n) = \frac{16n^2 - 3n^{\log_4 3}}{13}}$$

(b) Let $c = \log_4 3$. We prove that for all $n = 4^k$ $(k \geq 0)$,

$$T(n) = \frac{16n^2 - 3n^c}{13}.$$

Base case: $n = 4^0 = 1$

$$T(1) = 1, \qquad \frac{16 \cdot 1^2 - 3 \cdot 1^c}{13} = \frac{16 - 3}{13} = \frac{13}{13} = 1.$$

Holds.

Inductive hypothesis: Assume that for some $m = 4^\ell$ $(\ell \geq 0)$,

$$T(m) = \frac{16m^2 - 3m^c}{13}.$$

Inductive step: Consider $n = 4^{\ell+1} = 4m$. By the recurrence,

$$T(n) = 3T\left(\frac{n}{4}\right) + n^2 = 3T(m) + n^2.$$

Substitute the inductive hypothesis and $n = 4m$ (so $n^2 = 16m^2$):

$$\begin{aligned} T(n) &= 3 \cdot \frac{16m^2 - 3m^c}{13} + 16m^2 \\ &= \frac{48m^2 - 9m^c}{13} + 16m^2 \\ &= \frac{48m^2 - 9m^c + 208m^2}{13} \\ &= \frac{256m^2 - 9m^c}{13}. \end{aligned}$$

Now note that $n^c = (4m)^c = 4^c m^c = (4^{\log_4 3})m^c = 3m^c$, so $9m^c = 3 \cdot 3m^c = 3n^c$. Also $256m^2 = 16 \cdot 16m^2 = 16n^2$. Therefore

$$T(n) = \frac{16n^2 - -3n^c}{13},$$

which is exactly the formula for n. By the principle of mathematical induction, the closed-form holds for all $n = 4^k$, $k \geq 0$.

Problem 10: Recurrence relation and Induction [10 points]

A spider starts with 10 legs. Each day, its legs instantly double by splitting, but immediately after, exactly 2 legs fall off and are lost forever. Let a_n be the number of legs after n days, with $a_0 = 10$.

(i) **[2 points]** Write the recurrence relation for a_n.

(ii) **[4 points]** Find the closed-form formula for a_n using the expansion method.

(iii) **[4 points]** Prove that your closed-form formula is correct using mathematical induction.

Answer: (i) The recurrence relation is

$$a_n = 2a_{n-1} - 2 \quad (n \geq 1), \qquad a_0 = 10.$$

(ii) We use the recurrence $a_n = 2a_{n-1} - 2$ and substitute repeatedly:

$$\begin{aligned} a_n &= 2a_{n-1} - 2 \\ &= 2(2a_{n-2} - 2) - 2 = 2^2 a_{n-2} - 2 \cdot 2 - 2 \\ &= 2^2(2a_{n-3} - 2) - 2 \cdot 2 - 2 = 2^3 a_{n-3} - 2 \cdot 2^2 - 2 \cdot 2 - 2 \\ &\vdots \\ &= 2^n a_0 - 2(2^{n-1} + 2^{n-2} + \cdots + 2 + 1). \end{aligned}$$

The sum in parentheses is a geometric series:

$$2^{n-1} + 2^{n-2} + \cdots + 2 + 1 = 2^n - 1.$$

Thus,

$$a_n = 2^n a_0 - 2(2^n - 1) = 2^n \cdot 10 - 2 \cdot 2^n + 2 = 10 \cdot 2^n - 2^{n+1} + 2.$$

Simplifying:

$$a_n = 10 \cdot 2^n - 2^{n+1} + 2 = (10 - 2) \cdot 2^n + 2 = 8 \cdot 2^n + 2.$$

Therefore the explicit formula is

$$\boxed{a_n = 8 \cdot 2^n + 2 \quad \text{for all } n \geq 0}.$$

(iii) Proof by mathematical induction.

Base case ($n = 0$):

$$8 \cdot 2^0 + 2 = 8 \cdot 1 + 2 = 10 = a_0 \quad \checkmark$$

Inductive hypothesis: Assume that for some integer $k \geq 0$, $a_k = 8 \cdot 2^k + 2$.

Inductive step: Using the recurrence,

$$\begin{aligned} a_{k+1} &= 2a_k - 2 && (1)\\ &= 2(8 \cdot 2^k + 2) - 2 && (2)\\ &= 16 \cdot 2^k + 4 - 2 && (3)\\ &= 16 \cdot 2^k + 2 && (4)\\ &= 8 \cdot 2^{k+1} + 2. && (5) \end{aligned}$$

This is exactly the formula for $n = k + 1$.

By mathematical induction, $a_n = 8 \cdot 2^n + 2$ holds for all integers $n \geq 0$.

Problem 11: Graph [12 points]

(a) **[3 points, 1 each]** Determine whether the following statements are **True** or **False**. **No argument is needed**.

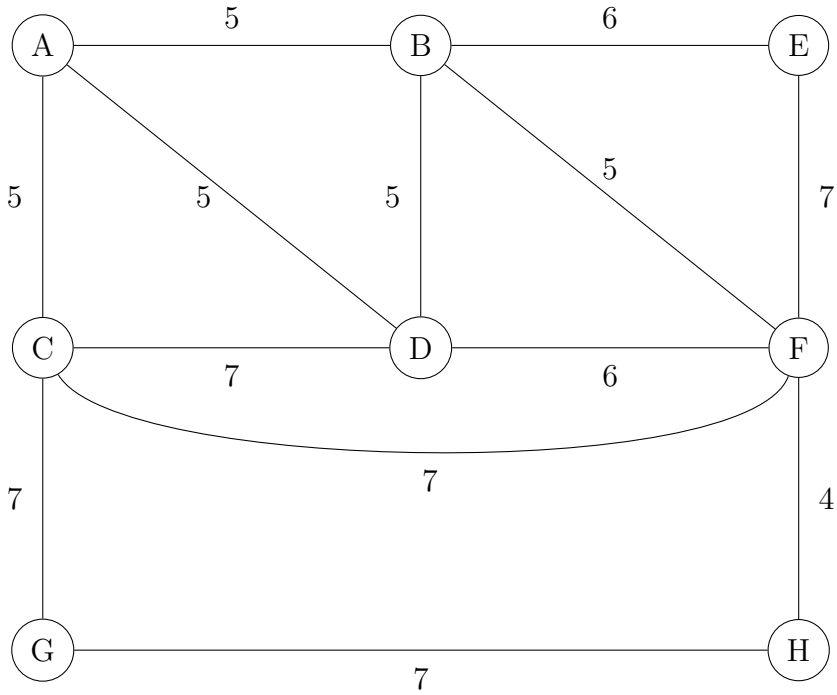
(i) A simple graph is bipartite if and only if it contains at least one odd cycle.

(ii) Consider a simple graph with 8 vertices and exactly 3 edges. This graph can have at least 2 and at most 5 isolated vertices. (An isolated vertex is a vertex of degree 0).

(iii) A simple graph with 10 vertices and 10 edges must contain a cycle.

(b) **[4 points]** Draw the simple graph G with 5 vertices and the following degrees $\{3, 3, 2, 1, 1\}$.

(c) **[5 points]** In the following graph, find the length of the shortest tour so that the postman can traverse every edge at least once and return to the starting point.

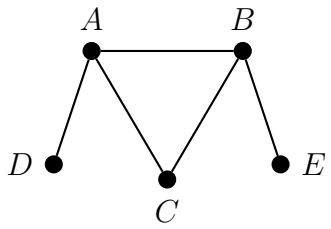


Answer: (a) (i) False

(ii) True

(iii) True

(b) Here is a simple graph with the following degrees.



(c) The graph is not Eulerian since two vertices have odd degrees (A, F). To make it Eulerian, we need to add the minimum weight of duplicate edges to pair up the odd-degree vertices. The shortest path between AF is $A \to B \to F$ with costs of 5+5.

The shortest tour is:

$$H \to G \to C \to F \to D \to C \to A \to D \to B \to A \to B \to F \to E \to B \to F \to H$$

Total weight calculation:

Total $= 7 \cdot 5 + 4 + 2 \cdot 6 + 5 \cdot 7 = 86$.

Thus, the shortest tour length is 86.

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